

CODEFEST 5

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} \text{ expected } C = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

$$1) \text{ } \&PE = \begin{bmatrix} w_{11}=5 & w_{12}=6 \\ w_{21}=7 & w_{22}=8 \end{bmatrix} \Rightarrow \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$2) t=0, \text{ input } a_{11}=1$$

$$a_{11}=1 \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \end{bmatrix}$$

$$t=1, \text{ input } a_{21}=3, a_{12}=2$$

$$a_{21}=3 \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 15 & 6 \\ 5+14 & \end{bmatrix}$$

$$a_{12}=2 \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 5+14 & 6 \end{bmatrix}$$

$$t=2, \text{ input } a_{22}=4$$

$$a_{22}=4 \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 18 & \\ 15+28 & 6+16 \end{bmatrix}$$

OUTPUT 19

$$t=3$$

$$\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 18+32 & \\ 43 & 22 \end{bmatrix}$$

OUTPUT

$$t=4$$

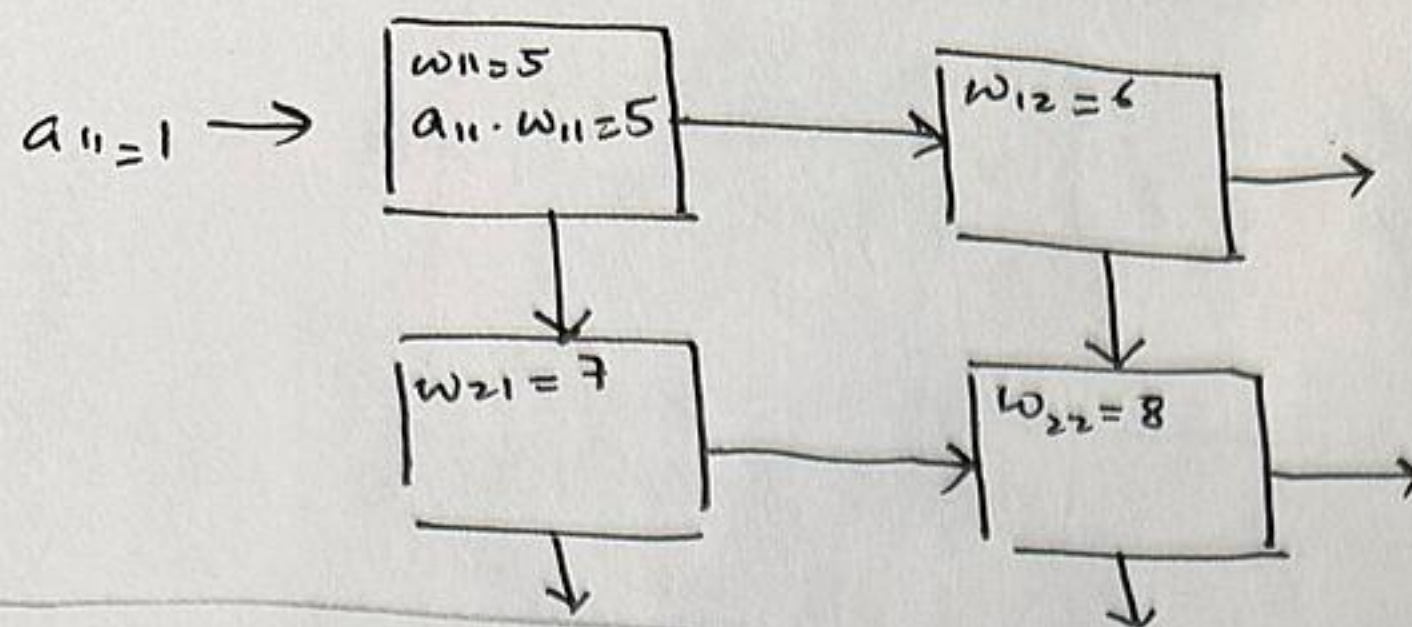
$$\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 43 & 50 \\ 19 & 22 \end{bmatrix}$$

OUTPUT

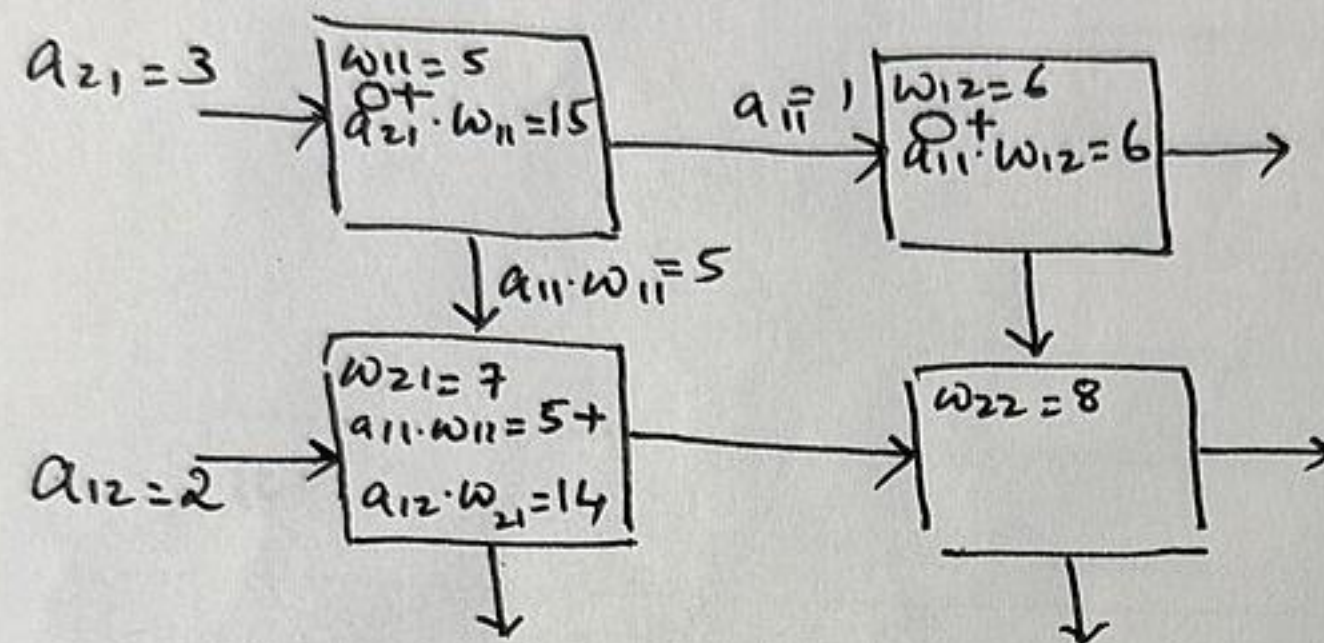
$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad PE = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

TIME	INPUT TO PE	PE VALUE	OUTPUT
$t=0$	a_{11} to p_{11}	$p_{11} = 5$	
$t=1$	a_{21} to p_{11} a_{11} to p_{12} (a_{12} to p_{21} + p_{11} to p_{21})	$p_{11} = 15$ $p_{12} = 6$ $p_{21} = 19$	
$t=2$	a_{21} to p_{12} (a_{22} to p_{21} + p_{11} to p_{21}) (a_{12} to p_{22} + p_{12} to p_{22})	$p_{12} = 18$ $p_{21} = 43$ $p_{22} = 22$	output = 19
$t=3$	(a_{22} to p_{22} + p_{21} to p_{22})	$p_{22} = 50$	output = 43, 22, 19
$t=4$			output = 50, 43, 22, 19

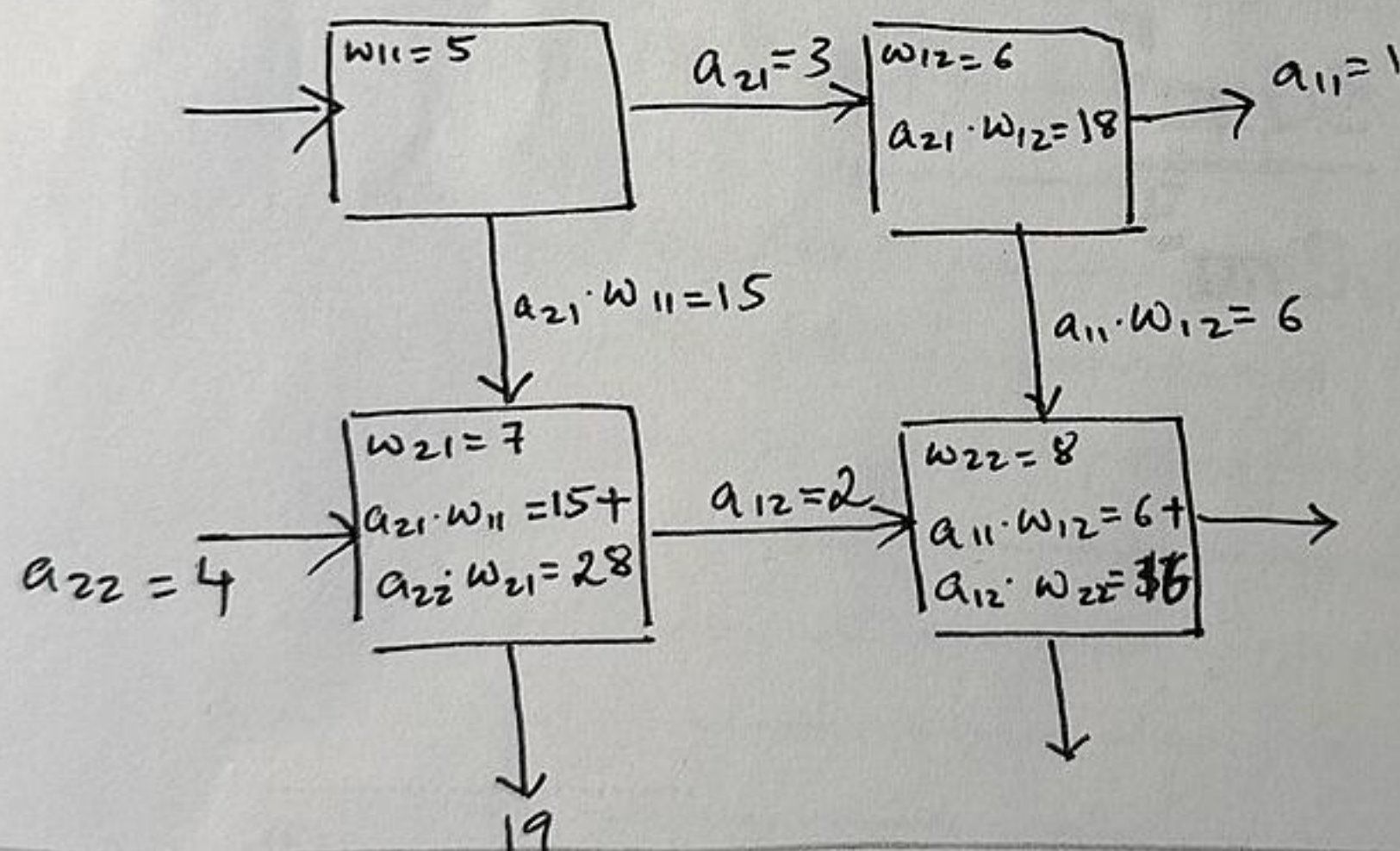
$t = 0$



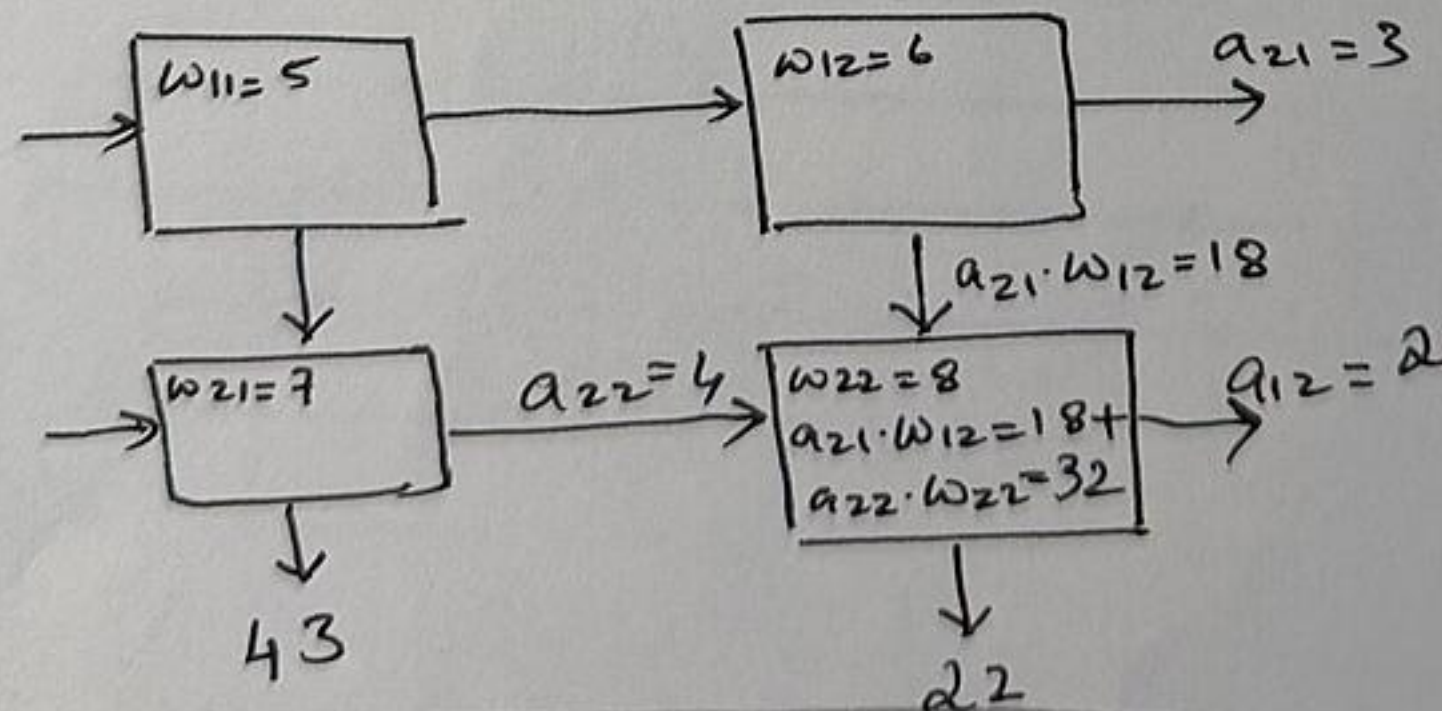
$t = 1$



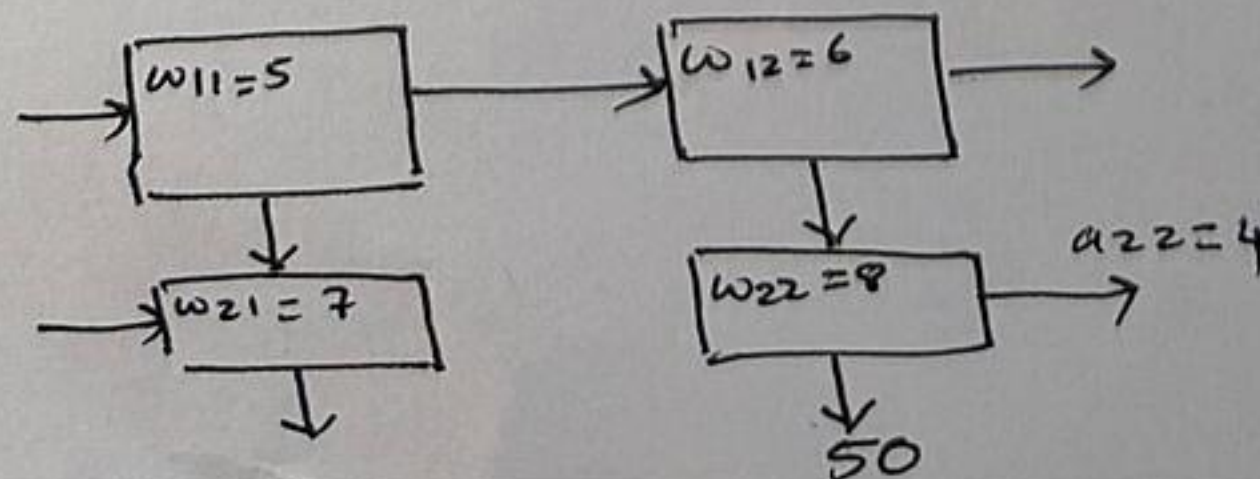
$t = 2$



$t = 3$



$t = 4$



3 a) Total MAC operations performed

$$N^3 = 2^3 \\ = 8$$

b) Number of times each input value is reused

Each input value is used twice.

For example, a_{11} is used with w_{11} and a_{11} is then passed to w_{12} .

c) Number of offchip memory accesses

For A and B (inputs)

Each element of A is read once $= N = 4$

Each element of B is preloaded as weight once $= N = 4$

For C (output write)

Each element of C is written once $= N = 4$

Total number of memory access

$$= N + N + N$$

$$= 4 + 4 + 4$$

$$= 12$$

d) In output-stationary each PE is responsible for each output element of the C matrix, thus, the A and B data flows and the partial computation remains stationary in the PE and do not move down the column.